Ahtsham Mustafa Awan

Geologist · Geospatial Data Scientist

https://portfolio-m1ie.vercel.app/ | 92 312 0343626 | Islamabad, Pak | ahtshamm214@gmail.com |
https://github.com/ahtshammustafa | https://www.linkedin.com/in/aawan1

Research Interests

Geoscience, Geospatial Artificial Intelligence, Machine/Deep Learning, Extreme Event Risk, Spatial Hydrology, Web GIS, Spatial Data Science

Skills

Programming & Databases: Python, R, C, JavaScript, SQL, ASP.NET, PostgreSQL/PostGIS

Remote Sensing / **GIS Software:** QGIS, ArcGIS Pro, ERDAS Imagine, ENVI, Google Earth Engine, GeoServer, Global Mapper, eCognition, GeoDa

Hydrological / **Hazard Modeling:** HEC-RAS, RAMMS, SCS-CN/AMC modeling, wflow, SWAT, geomorphometric flood-risk ranking, pysheds terrain analysis

GeoAI: PyTorch, U-Net, ResNet, DenseNet, GeoAI, EfficientNet, ViT, MaxEnt, AHP/Fuzzy Logic MCDA

Web GIS React, Node.js, Leaflet.js, Openlayers, ArcGIS Maps SDK, MapBox, Next Js, ArcGIS Enterprise

Remote Sensing Datasets Sentinel-1, Sentinel-2, Landsat, MODIS, ASTER GDEM, SPOT, Pleiades, PRSS, RADAR, LIDAR, Maxar, Planet Labs, CHIRPS, GPM, TRMM, USDA soil data, SEN12-FLOOD, Sen1Floods11

Geotechnical Software: Slide2, Plaxis 2D/3D, GeoStudio (SLOPE/W, SEEP/W), RocFall, Dips, Swedge, RS2, RockWorks

Projects & Research

SAR–Optical Flood Segmentation Framework - 2025 Pakistan floods

– A pixel-level flood segmentation architecture combining U-Net decoders with ResNet-50, DenseNet-121, and EfficientNet-B0 encoders, fusing Sentinel-1 SAR, Sentinel-2 optical imagery, and DEM-derived hydrogeomorphological indices (slope, TWI, HAND). Terrain preprocessing is built around pysheds and aligned to the Sen1Floods11 benchmark.

Multi-Sensor Flood Scene Classification (SEN12-FLOOD) - 2022 Pakistan floods

– Fine-tuned four deep learning backbones (ResNet-50, DenseNet-121, EfficientNet-B0, ViT) on fused SAR/optical imagery for flood scene classification, then combined them through soft-voting and stacking ensembles (logistic regression, XGBoost, SVM), reaching 95.6% test accuracy.

Landslide Susceptibility Mapping, Azad Kashmir

– Utilised GeoAI and Deep Learning architectures (U-Net, ResNet, ConvLSTM) alongside spatial multicriteria decision analysis (AHP and Fuzzy Logic) to analyze geotechnical testing data (Atterberg, Casagrande, direct shear, triaxial, sieve) and multi-source Earth observation data (Sentinel, Maxar, Planet Labs, Pleiades) for landslide susceptibility mapping. Also, executed repeat-pass InSAR time-series analysis to identify micro-topography deformation, trace slope acceleration boundaries, and quantify active landslide displacements across high-risk corridors in Azad Kashmir.

Mangla Watershed Runoff & AMC Modeling

– Developed a Google Earth Engine algorithm fusing MODIS LULC and USDA soil data to automate AMC maps and SCS-CN hydrological modeling, analyzing 2005-2015 runoff for the Mangla watershed. Validated CHIRPS, GPM, and TRMM outputs against ground metrics, identifying CHIRPS as achieving up to 90% accuracy.

Jhelum River Basin Flood Susceptibility Mapping

– Modelled Jhelum River Basin flood susceptibility using ASTER GDEM data in ArcMap by employing an optimized Geo-Morphometric ranking approach. Evaluated how basin gradient, relief, bedrock geology, and sub-basin area drive flash and riverine flood risks to inform government risk reduction strategies.

Pakistan Flood Monitoring Dashboard

– Led in the development of a full-stack Pakistan Flood Monitoring Dashboard using React, Leaflet.js, and Node.js

to ingest real-time WAPDA telemetry and automate Sentinel-1 SAR satellite raster processing. Architectured scalable PostgreSQL/PostGIS databases and GeoServer pipelines, reducing geospatial query latency while streaming live driver discharge analytics and emergency alerts.

Badswat Glacier Debris Flow Hazard Assessment

– Modelled and simulated Badswat Glacier debris flow hazard assessment using the RAMMS model and HEC-RAS.

Water Quality Modeling Using Landsat 8 & Hyperspectral Data

– Modeled water quality parameters using Landsat 8 and ASD hyperspectral spectroradiometer data.

Coastal Inundation Vulnerability, Indus River Delta

– Simulated coastal inundation vulnerability and impact assessment for the Indus River Delta using HEC-RAS.

Built Environment & Extreme Event Risk

Himalayan Landscape Dynamics: LULC & LST Analysis (2002-2026)

– Analyzed twenty four years (2002-2026) of Himalayan landscape dynamics by deploying SVM classification on Landsat imagery to map LULC changes against split-window algorithm retrieved Land Surface Temperatures. Modeled strong LST-NDVI correlations (R^2 up to 0.87) to quantify glacier decline (6.37%) and barren expansion (6.47%) for climate resilience planning.

Kohat City Urban LULC & LST Analysis (1998-2018)

– Analyzed Land Use Land Cover (LULC) dynamics and Land Surface Temperature (LST) for Kohat City, Pakistan, using multi-temporal Landsat imagery from 1998 to 2018. Quantified a 55% built-up expansion and a $4.6°C$ increase in mean LST, demonstrating the direct impact of urban growth on microclimate changes.

Peshawar Groundwater Quality Assessment

– Evaluated the impact of urbanization and agricultural land-use changes on Peshawar's groundwater by classifying multi-temporal Landsat images and analyzing 217 tube well samples across 7 parameters. Correlated anthropogenic pollution with declining water quality and rising health risks, documenting a 14% increase in methemoglobinemia alongside elevated cases of UTIs and peptic ulcers.

Aedes Albopictus Distribution Modeling, Rawalpindi

– Constructed high-resolution bioclimatic and LULC datasets from satellite data to model Aedes albopictus mosquito distribution in Rawalpindi using the MaxEnt algorithm. Identified roadside green patches, winter temperatures, and household waste practices as primary risk factors, incorporating community surveys to guide targeted vector control.

Education

Master of Science in Remote Sensing and Geographical Information Systems 09/2021 - 07/2024

National University of Sciences and Technology, Islamabad

Final grade 3.55/4.0. Thesis: Comparison of different Machine Learning Algorithms for Land Use Land Cover Change Detection and its Relationship with Land Surface Temperature and Topography. Relevant coursework: Spatial Analysis and Modeling, Advanced Remote Sensing and Digital Image Processing, Advanced Geodatabase and Programming, Natural Hazards and Disaster Management.

BS Applied Geology 10/2016 - 02/2021

The University of Azad Jammu and Kashmir, Muzaffarabad

Final grade 3.54/4.0. Thesis: Landslide Mapping and Spatial Distribution Analysis from Muzaffarabad to Luat, Neelum Valley, Azad Kashmir, Pakistan. Relevant coursework: Economic Geology, Petrology, Mineralogy I & II, Geochemistry, Structural Geology, Geophysics, Sedimentation & Sedimentary Basins of Pakistan.

Work Experience

Assistant Research Officer 01/2024 – Current

Government of Pakistan - Ministry of Defense

– Numerical modeling for disaster management, photogrammetry, slope mapping, terrain analysis, and infrastructure mapping using diverse datasets, including SPOT, Pleiades, PRSS, RADAR, and LIDAR. Conducted drones and land surveys for advanced hydrological modelling and post landslides and floods impact assessment. Led in

the development of state of the art web applications (Flood Monitoring, Spatial Data Analytics, Object Detection and AI Agents Dashboards) for geospatial data analysis and visualization.

Research Assistant 01/2023 – 12/2023

National University of Sciences and Technology, Islamabad

– Mapped extreme weather events (Floods, Landslides and Heatwaves) using Remote Sensing and Artificial Intelligence Techniques. Performed numerical modeling and simulations for coastal inundation vulnerability and flood hazard evaluations.

Remote Sensing and GIS Intern 01/2021 – 08/2021

Pakistan Council of Research in Water Resources

– Identification of Groundwater Potential Zone using GIS/Remote Sensing Techniques and AHP Conducted water quality testing, surveying, and mapping for environmental assessments. Prepared comprehensive reports and documentation based on survey findings and analysis.

Field Excursion

Geological & Geotechnical Field Investigations

– Conducted extensive field mapping and geotechnical investigations across the Salt Range (Eastern, Central, and Western regions) for structural analysis, stratigraphic profiling, and slope stability assessment.
– Performed flood impact assessment and debris flow hazard analysis in Northern Pakistan, including Azad Kashmir, Gilgit Baltistan, and Chitral, identifying landslide triggering factors such as lithology, slope gradient, rainfall intensity, and seismic activity.
– Executed geotechnical engineering tests including direct shear test, triaxial shear test, unconfined compression test, uniaxial compressive strength (UCS), point load test, Brazilian tensile strength test, Atterberg limits (liquid limit, plastic limit, shrinkage limit), Casagrande plasticity chart classification, and sieve analysis for grain size distribution.
– Utilized advanced software tools for slope mapping and geotechnical analysis including Slide2 (Rocscience), Plaxis 2D/3D, GeoStudio (SLOPE/W, SEEP/W), RocFall, Dips, Swedge, RS2, and RockWorks for kinematic analysis, slope stability modeling, and rockfall simulation.
– Conducted detailed landslide susceptibility mapping, impact analysis, and risk assessment using integrated field data and remote sensing techniques.
– Prepared comprehensive geological maps, cross-sections, and structural contour diagrams for various terrains, contributing to disaster risk reduction strategies.

Certifications

Online Certifications | MOOCs | Trainings

– Satellite Imagery, Remote Sensing & Machine Learning Specialization (Coursera)
– IBM AI Engineering (Coursera)
– Machine Learning Specialization
– Deep Learning Specialization
– Image and Video Processing: From Mars to Hollywood with a Stop at the Hospital - Duke University
– Fundamentals of Digital Image and Video Processing (Coursera)
– Spatial Data Science and Applications - Yonsei University (Coursera)
– Spatial Data Science: The New Frontier in Analytics - ESRI Training (MOOC)
– Hyperspectral Thermal Image Unmixing - Geo University
– GIS for Climate Action (MOOC) - ESRI
– Imagery in Action (MOOC) - ESRI
– Active and Passive Satellite Data Analysis Using Cloud Computing for Surface Water/Flood Mapping

Languages & Activities

Urdu - Native, **English** - C1 (TOEFL)

Social and Multicultural: Co-founder, Hamdard Welfare Foundation Azad Jammu and Kashmir; Member, Alkhidmat Foundation Pakistan; Climate Change Activist focusing on Multi-Hazards in Pakistan; Member, NUST Environment Club.

Recommendations

Dr. Javed Iqbal — MS Thesis Supervisor

Institute of Geographical Information Systems, NUST
javed@igis.nust.edu.pk (+92) 90854473

Dr. Ejaz Hussain — Associate Dean and MS Co-Supervisor
Institute of Geographical Information Systems, NUST
ejaz@igis.nust.edu.pk (+92) 5190854400

Dr. Salman Atif — Associate Professor | Instructor
Institute of Geographical Information Systems, NUST
salman@igis.nust.edu.pk

Dr. Ali Wahid — Assistant Professor | Instructor
Institute of Geology, University of Azad Jammu and Kashmir
ali.wahid@live.com